A SUBCOLLECTION OF OPENS SATISFYING A COVERING AND INTERSECTION CONDITION DEFINES A QUOTIENT-LIKE MAP TO A SPACE WITH THOSE OPENS AS A BASIS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-topology/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The sufficient condition for a collection of subsets to be a basis. Invoked inside the counted proof (source lines 616-618). Provides the unique topology on the quotient set of the equivalence relation.

Source reference. Original lemma lemma-make-base: sources/stacks-topology/pinned-original-source.tex, source lines 515–515. Navigation pointer only: the counted text refers to the Stacks item lemma-make-base, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let X be a topological space. Let $\mathcal{B}$ be a collection of opens of X. Assume $X = \bigcup_{U \in \mathcal{B}} U$ and for $U, V \in \mathcal{B}$ we have $U \cap V = \bigcup_{W \in \mathcal{B}, W \subset U \cap V} W$. Then there is a continuous map $f : X \to Y$ of topological spaces such that*

(1) *for $U \in \mathcal{B}$ the image $f(U)$ is open,*
(2) *for $U \in \mathcal{B}$ we have $f^{-1}(f(U)) = U$, and*
(3) *the opens $f(U)$, $U \in \mathcal{B}$ form a basis for the topology on Y.*

Proof. Define an equivalence relation $\sim$ on points of X by the rule

$$x \sim y \Leftrightarrow (\forall U \in \mathcal{B} : x \in U \Leftrightarrow y \in U)$$

Let Y be the set of equivalence classes and $f : X \to Y$ the natural map. Part (2) holds by construction. The assumptions on $\mathcal{B}$ exactly mirror the assumptions

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in Lemma lemma-make-base on the set of subsets $f(U)$, $U \in \mathcal{B}$. Hence there is a unique topology on Y such that (3) holds. Then (1) is clear as well. $\square$